

Contains-Matching Best Practices

Making the Most of your ISP Watch List Entries

November 2020



Who is Using Contains-Matching?

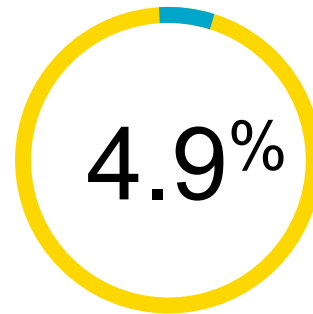
In **June 2020** the watch list contains-matching feature, also called *fuzzy matching*, was released to production.

This feature allows customers to target a partial value from a watch list instead of the exact value. The watch list rule would fire when that partial value appears anywhere in the field targeted.

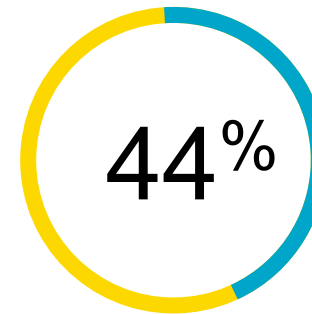
Any eligible watch list can utilize contains-matching simply by changing the match-type of the individual entry from “exact” match to “contains” match.

Feature Adoption Metrics

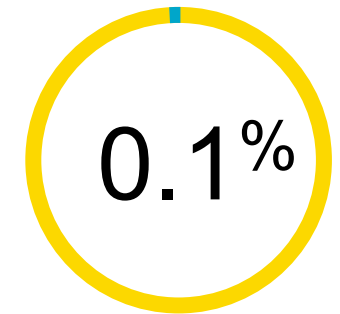
As of 11/5/2020



31 out of 633
active subscribers using Device Risk services have a CONTAINS match entry in an existing watch list.



12 out of 27
eligible contains-match watch list types are actually in use. Browser Version and ISP are the most popular.



42 out of 37,410
existing ISP watch list entries use a CONTAINS match type instead of EXACT match type.

How can we raise awareness?



Re-Education

- Use this deck to explain how it works and provide Best Practice Recommendations on use.



Relevant Examples

- Provide each subscriber a spreadsheet of concrete examples that convert their existing watch list members from EXACT values to a CONTAINS alternative.



Actionable Data

- Prove the value of using contains matching by quantifying the increased fraud catch for each alternative value provided.



Ongoing Engagement

- Use the **Potential Watchlist Entries Report** to identify future device or transaction attributes that present elevated fraud risk. Apply Best Practices to chosen entries.

Information Available Today

BI has prepared files for you to use when re-educating and engaging your customers to promoting this feature.

All files are available in the Customers Success shared Google Team drive:

Google Drive
>> Shared Drives
 >> Customer Success
 >> Business Intelligence
 >> Fuzzy Matching Best Practices
 >> *your files*



ISP Best Practices

Use this presentation to explain how the feature works and share best practices to consider. Slides from here forward are for the client presentation.



Utilize your BI SMEs as a guest presenter in your upcoming client call, if desired.



Client Recommendations

Each client manager has a spreadsheet named for them. This sheet contains concrete ISP contains match alternative values that can be used.



General Support clients are not eligible for this professional service.



Supplemental Data

Share the printed slide deck and list of common ISP suffixes to compliment the actionable data the client can use for further analysis and future decisioning.



Common ISP suffixes can be removed from watch list entries to increase fraud catch.

Level-set on Functionality

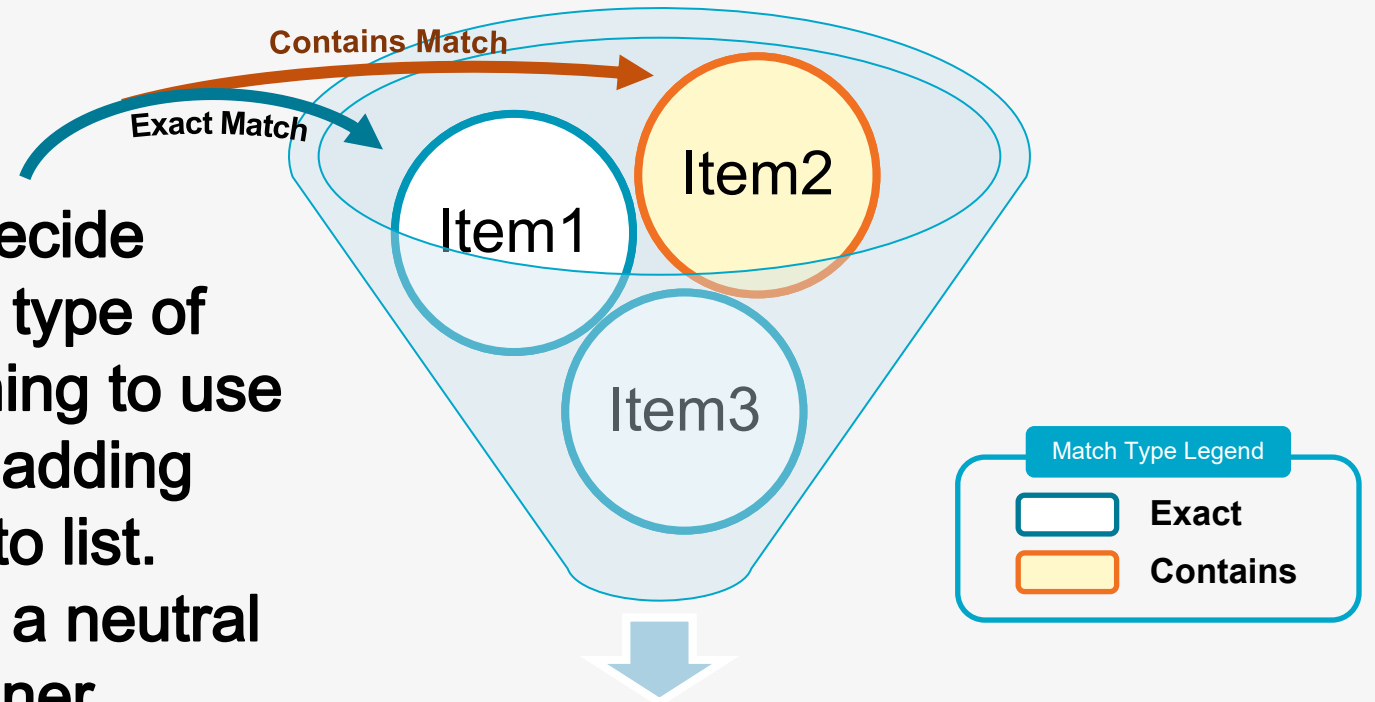
Any existing watch list can utilize fuzzy matching simply by changing the match-type of the individual entry from “exact” match to “contains” match.

Any of the following watch lists can incorporate this fuzzy-matching feature:

- Application Name
- Application Version
- Browser Type
- Browser Version
- Carrier Name
- City
- Device Brand Name
- Email Address
- Email Domain
- Event ID
- ISP Name
- ISP Organization
- Language
- Manufacturer
- Online ID
- Operating System
- Postal Code
- Region
- Screen Resolution
- Security PIN
- Security PIN Type
- Service ID
- Session Alias
- SKU
- Street
- Tenant ID
- URL

You decide which type of matching to use when adding entry to list.
List is a neutral container.

How Contains-Match Really Works



Your Watch List

Level-set on Functionality

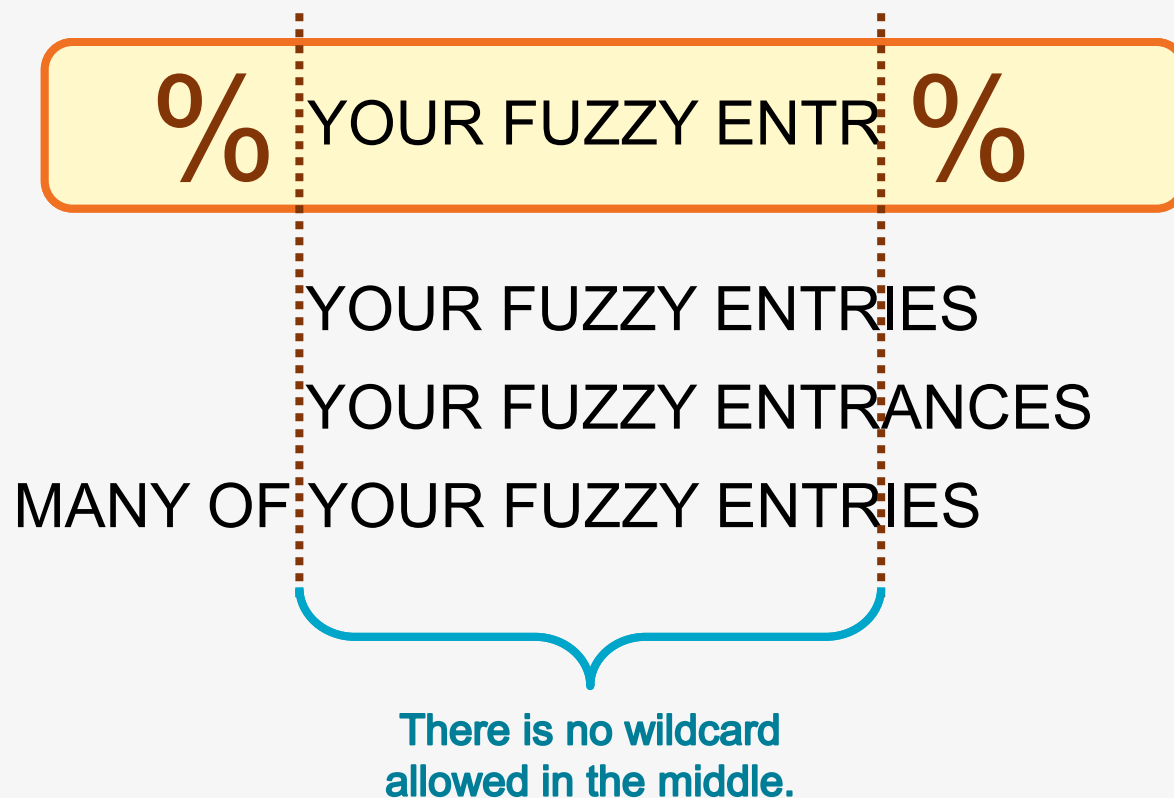
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- Screen Resolution
- Security PIN
- Security PIN Type
- Service ID
- Session Alias
- SKU
- Street
- Tenant ID
- URL

Wildcards (%) are assumed at the start and end of your entry when comparing.

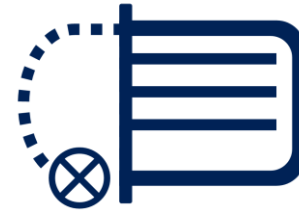
How Contains-Match Really Works



Remove Leading Stop Words

Remove the first word in the service provider name when it is a descriptor word such as “THE”, “A”, “AN”. These articles of grammar simply describe a term in the primary business name but do not differentiate that entity from another. Removing those starting descriptors from the name will improve your fraud catch when the registered business name has a slight variation over time or in a different region.

** Ensure there is a space at the end of the stop word as a guide for deciding whether or not to remove that word.*



Exact Match Examples:

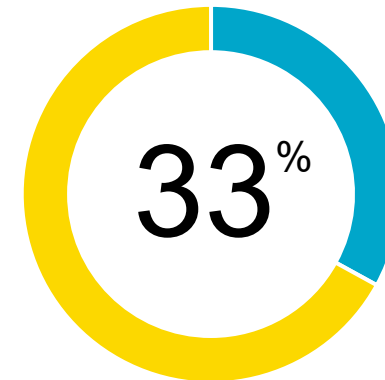
A COCA-COLA COMPANY
THE COCA-COLA COMPANY
THE ENDURANCE INTERNATIONAL GROUP
THE OPTIMAL LINK CORPORATION
AN INTERNET SERVICE PROVIDER IN PAKISTAN

Improved Fuzzy Match Entries:

COCA-COLA COMPANY
 ENDURANCE INTERNATIONAL GROUP
 OPTIMAL LINK CORPORATION
 INTERNET SERVICE PROVIDER IN PAKISTAN

Reminders & Cautions

- **Avoid blanket removal of the characters** without considering whether or not they form a word or is simply the start of a word.
 — THEATER ABCD would not be changed.
- Look for a **space separating the leading article of grammar** from the rest of the name.
- Remember to **remove trailing and leading spaces** after the stop word is removed.
- Be mindful of the language of the provider to look for **stop words outside of the US English** examples referenced here.
 — Common Spanish Stop Words: DE, LA, EL, Y, etc.



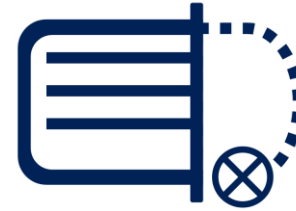
of subscribers with an ISP watch list have an existing entry that would strategically target more transactions once the leading stop words were removed from the ISP name.

+1,187 TXNs
 in recent 3 months of activity.
 (155 out of 458 Subscribers)

Remove Common Name Extensions

Remove common business name extensions that reside at the end of your service provider name. These suffixes refer to the same entity but create enough variation to thwart rules focused upon exact names. To employ fuzzy matching, remove common extensions provided by the Customer Analytics team* before adding that entry to your watchlist. This list includes various alternatives that punctuation changes applied that may limit fraud catch when exact matching is applied.

**See the appendix for the list of common extensions to remove.*



Exact Match Examples:

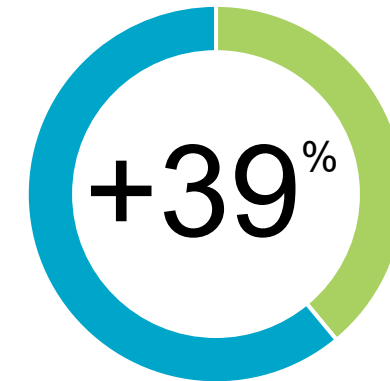
THEATER POR CABLE **S. R. L.**
 SMARTLINK SOLUTIONS **PVT. LTD.**
 WANTEL TECNOLOGIA **LTDA. EPP**
 1&1 INTERNET **AGI**

Improved Fuzzy Match Entries:

THEATER POR CABLE
 SMARTLINK SOLUTIONS
 WANTEL TECNOLOGIA
 1&1 INTERNET

Reminders & Cautions

- Some service providers **change the extension or add punctuation to create a new name** that bypasses your existing exact-match watchlist entry.
- Be sure to look for names that include a **combination of extensions** stacked at the end. Remove combinations as well.
- If your customer base is primarily outside of the US, there **may be additional punctuation variations** not referenced in the appendix.
- Remember to **remove trailing spaces** after the suffix is removed.



more transactions are targeted using contains matching after the common extension was removed from name of the existing ISP watch list entry.

+1.2K Avg per WL Entry

(+24 million transactions)

Remove Location Reference Near/At End of Provider Name

When the business name has a country, continent, or region reference near the end of the name, then that primary business entity is likely operating from multiple locations, and changes to that geolocation reference will prevent you from targeting all traffic from the primary entity. Remove that reference to target any future relocations of your risky ISPs. Remember that once you remove the common extension (previous slide), the choice to remove the location reference is clearer.



Exact Match Examples:

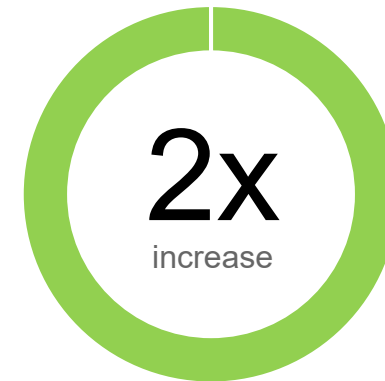
VOLKSWAGEN GROUP **OF AMERICA**
 MAGNA CAPAX **FINLAND OY**
 HOST1PLUS HOSTING SERVICES. **BRAZIL.**
 ARDH GLOBAL **INDONESIA PT**
 UBIQUITY SERVER SOLUTIONS **SEATTLE**

Improved Fuzzy Match Entries:

VOLKSWAGEN GROUP
 TOYOTA TSUSHO
 HOST1PLUS HOSTING SERVICES
 ARDH GLOBAL
 UBIQUITY SERVER SOLUTIONS

Reminders & Cautions

- Remember to remove dangling particles like OF, IN, DE, and DES that remain after the location is removed.
- Exceptions:** Do not remove that location reference when the name starts with one of the entity types below, which commonly uses a geolocation reference but variations do not represent the same business entity.
 - UNIVERSITY OF...
 - BANK OF...
 - STATE OF...
 - COLLEGE OF...
 - CITY OF...
 - PORT OF...
- Do not remove the location references that appear at the start of the name.



FRAUD CATCH

when an existing ISP watch list entry was modified to remove location references at or near the end of the ISP name

Remove Leading House Number or Trailing Unit Number from Address

When the service provider name fits the format of a street address, that house number at the start of the name or unit number at the end can easily be changed to create a new ISP name that thwarts exact matching rules. Fraudsters frequently make small modifications to a fictitious business name or street address to circumvent current barriers created by watch lists using exact matching to that business name.



Exact Match Examples:

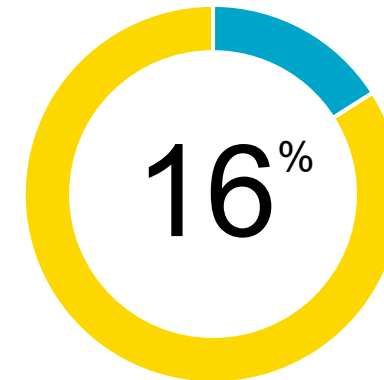
3100 CUMBERLAND BLVD ATLANTA GA
 150 CECIL STREET #14-03
 60 MARKET SQUARE P.O. BOX 364
 EQUIFAX TECHNOLOGY (IRELAND) LIMITED
 - 170802750

Improved Fuzzy Match Entries:

CUMBERLAND BLVD ATLANTA GA
 CECIL STREET
 MARKET SQUARE P.O. BOX
 EQUIFAX TECHNOLOGY (IRELAND) LIMITED

Reminders & Cautions

- Remember to remove dangling special characters like dashes (-), periods (.), or hashtags (#) that remain after the location is removed.
- Exceptions:** Do not remove that leading/trailing numbers when the remaining name for contains matching leaves a single word, is short, or is a generic term commonly used in online transactions.
 - CLOUD 9 LTD. >> CLOUD
 - E-COMM 911 >> E-COMM
 - DATA 102 LLC >> DATA
- Target contains matching for those ISP names that are specifically formatted as a street address.



of existing ISP watch lists contain an entry that is formatted as a street address and have a leading or trailing number that would strategically target more transactions once the ending numbers were removed from the name.
 (196 out of 1251 watch lists)

1 Remove Leading Stop Words

2 Remove Common Name Extensions

3 Remove Location Reference Near/At End of Provider Name

4 Remove Leading House Number or Trailing Unit Number from Address

Best Practices in Action

Applying the Logic Provided to Example
ISP Watch List Entries

1&1 VERSATEL DEUTSCHLAND GMBH

HOSTINGER INTERNATIONAL LIMITED

60 MARKET SQUARE P.O. BOX 364

CANTV SERVICIOS, VENEZUELA

NAZWA.PL SP.Z.O.O.

HUBEI YUNWUTONG NETWORK TECHNOLOGY CO. LTD.

GATEWAY SYSTEMS (INDIA) PVT. LTD., PUNE.

Avoid creating a contains entry that is a single, common word like “INTERNET” because it will target too many providers and increase false positives.

Exceptions to Combining All Best Practices

Examining Cases where the combined set of Best Practices are Too Fuzzy

These spelling variations may tempt you remove too many characters and leave an entry that is too short and generic.

1&1 INTERNET
1 & 1 INTERNET

Removing the extension and country references here leaves words that are common terminology in web hosting and data processing.

DATACENTER D.O.O.

INTERNET SERVICE PROVIDER IN PAKISTAN

Be mindful of the language to ensure the country is not removed from an item on the exceptions list.
UNIVERSITY = UNIVERSIDAD

UNIVERSIDAD POLITECNICA
DE MADRID

General Reminders

Making the Most of your Contains-Match Watch List Entry



Ensure there are no trailing or leading spaces in the watchlist entry when using copy/paste to populate your list. Nonprintable characters still change the value of your watchlist entry, even when you cannot see them.



Be mindful of instances where the ISP / ISPORG is referencing an aggregator or actual subscriber. These names should not be included in watchlists targeting negative behaviors.

- Aggregator Examples: AKAMAI, etc.
- Contact your Customer Success Manager or email clientsupport@iovation.com if you have questions about any ISP names referencing an existing subscriber.



Be mindful of the language of your customers' local ISP / ISPORG. The best practices provided reflect English language terms but the region you serve may have a different base language and naming rules.

- Spanish Stop Words: DE, Y, EL, LA



Do not include the ISP / ISPORG of major mobile carriers in your contains-match watchlist entry.

- You may inadvertently create friction or block a large number of customers from interacting with your website, mobile app, or brand.



Request the Potential Watchlist Entries Report for a list of recent ISP / ISPORG values seen in your traffic.

- Contact your Customer Support Manager, if you have dedicated support, or your account manager to request this report and review new watchlist recommendations.

Next Steps



Review the list of contains-match alternative ISP names for your existing watch list entries.



Verify the increased fraud catch and manual review volume are acceptable for preferred alternates.



Add preferred alternates to your existing ISP watch list(s) as contains match entry.

FAQs

Watch List Tuning Frequently Asked Questions

ISP & ISPORG Best Practices



Watch List Tuning FAQs:

How can I find out what ISPs or ISP Organizations are risky enough to warrant inclusion in a watch list?

Request the Potential Watchlist Items Report from your Customer Support Manager or Account Manager (for general support customers) to view the list of ISPs seen in the past 90 days that present some level of risk. This report narrows the vast ISPs experienced to only those with a transaction volume high enough to warrant targeting with a rule:

- (1) ISP has at least 1 account or device with confirmed fraud or abuse;
- (2) Client has seen as least 5 transactions in past 3 months from that ISP;
- (3) At least 1 business rule triggered in the past 3 months; or
- (4) Transactions from that ISP have returned a REVIEW or DENY response at some point during that window.

This report also shows risky screen resolutions, browsers, and 5+ other commonly-targeted risky transaction/device attributes.

Where can I get a list of the ISPs and ISP Organization currently in my watch list?

You can use the Intelligence Center to view all watch list entries or request a custom report from your Customer Support Manager or Account Manager (for general support customers). There may be a fee for the report, depending on your current Professional Services support package.

Watch List Tuning FAQs:

How do I know if an ISP in the provided report is *preferred* and merits inclusion in a watch list?

In the Potential Watchlist Entries Report, use the existing priority ordering of the report to examine the most risky ISPs first. If the entry is already included in your watch list, it will be flagged in the *On Existing Subscriber Watchlist* column and the row will be highlighted in blue. For those items NOT already in your watchlist, consider both the rate of fraud associated with that ISP (**FRAUD_RATE**) and the rate of DENY transactions (**DENY_RATE**).

When those two metrics are not sufficient factors for determining risk, also consider using the no-device rate metrics and behavioral columns to the far right in the report. Those behavioral columns outline additional risk factors that occurred in traffic from the current ISP (or other transaction/device attribute under consideration).

How do I search in Excel to find rows in my report that might trigger on my new contains-match entry?

For contains matching, use the column filter feature in Excel to see what additional ISPs will be captured when the name partially contains the value you are considering for your fuzzy-match watch list entry.